



UNIVERSITY OF WEST ATTICA
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING AND
COMPUTERS

LABORATORY
SIGNAL & SYSTEMS

LABORATORY EXERCISE #3 –SYSTEMS
CONTINUOUS TENSE

SCHEDULE YEAR 2018-2019

LABORATORY EXERCISE 3

CONTINUOUS TIME SYSTEMS

Continuous-time signals or analog signals are signals whose independent variable varies over a continuous interval of values. In one-dimensional signals the domain of the signal is the interval of the real number line.

A system is defined as an entity that, acting on a signal $x(t)$, results in another signal $y(t)$. From a mathematical point of view, a system can be considered as a transformation $\mathcal{Y}$ that transforms one signal into another signal.

$y(t) = \mathcal{Y}\{x(t)\}$. More simply we can say that a system receives (or is excited by) an input (or an input signal) $x(t)$ and produces an output (or an output signal) $y(t)$.

System categories

In this section we will classify systems into categories according to some specific criteria.

1. Categorization according to the number of inputs and outputs

The first distinction between system categories depends on the number of allowed inputs and outputs.

- Single-Input, Single-Output or SISO (Single-Input, Single-Output) systems.

An example of a simple one-input - one-output system is a system described by an input-output relationship of the form $y(t) = ax(t)$ (where a is a scalar quantity, t is a time system, and $x(t)$ is the input to the system, $y(t)$ is the output to the system).

So let's say a system governed by the input-output relationship $y(t) = 2x(t)$ this system is excited by an input $x(t) = \cos(t)$. You are asked to draw the output of the system. The

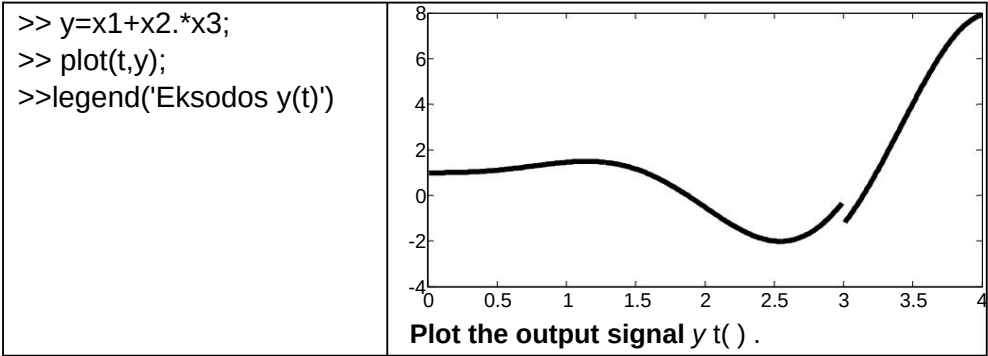
COMMANDS	RESULT	COMMENTS
<pre>>> t=0:.01:3; >> x=t.*cos(t); >> plot(t,x); >>legend('Eisodos x(t)')</pre>		Planning of signal entrance $x(t)$.
<pre>>> plot(t+2,x); >>legend('Eksodos y(t)')</pre>		Planning of signal exit $y(t)$.

• Multiple-input, single-output systems or MISO systems (Multi-Input, Single-Output).

Let us therefore consider a system governed by the input-output relationship $y(t) = \sin(t)$. This system is excited by inputs $x_1(t) = \cos(t)$, $x_2(t) = t$, and $x_3(t) = t^2$.

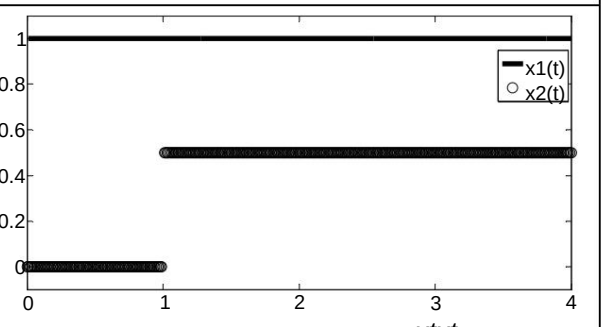
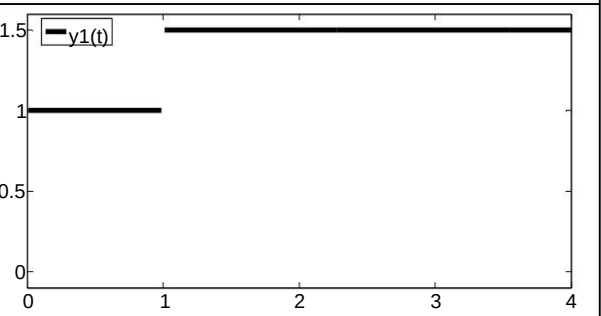
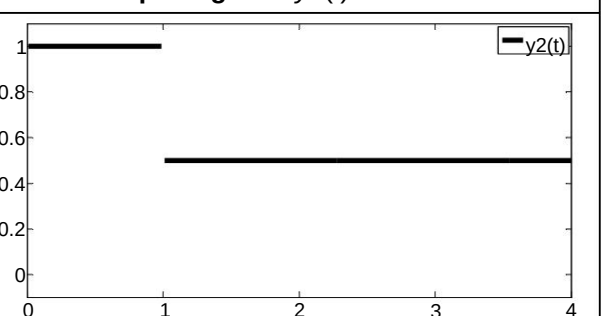
is excited by inputs $x_1(t) = \cos(t)$, $x_2(t) = t$, and $x_3(t) = t^2$. You are asked to draw the output $y(t)$ of the system.

COMMANDS	RESULT-COMMENTS
<pre>>> t=0:.01:4; >>x1=heaviside(t)- heaviside(t-3); >> x2=t.*sin(t); >> x3=t.*cos(t); >> plot(t,x1,t,x2,t,x3); >> legend('x1(t)','x2(t)','x3(t)')</pre>	Design of input signals $x_1(t)$, $x_2(t)$, $x_3(t)$.



• Systems with multiple inputs and multiple outputs, known as MIMO (Multi-Input, Multi-Output) systems. An example of such a system is a system $\ddot{y} = \ddot{y}$ and described by an of the form, $y(t) = x_1(t) + x_2(t)$ input-output relationship

$y(t) = x_1(t) + x_2(t)$
So let us consider a system governed by the input-output relationship $\ddot{y} = \ddot{y}$
 $y(t) = x_1(t) + x_2(t)$ and $y(t) = x_1(t) + x_2(t)$. This system is stimulated by system $x_1(t) = 1$, $x_2(t) = 0.5 \cdot 1$. You are asked to plot the output of inputs in a suitable time interval.

COMMANDS	RESULT - COMMENTS
<pre>>> x1=heaviside(t); >> x2=0.5*heaviside(t-1); >> plot(t,x1,t,x2,'o'); >> ylim([-0.1 1.1]); >> legend('x1(t)','x2(t)')</pre>	 Design of input signals $x_1(t)$, $x_2(t)$.
<pre>>> y1=x1+x2; >> plot(t,y1) >> ylim([-0.1 1.6]); >> legend('y1(t)')</pre>	 Plot the output signal 1 $y(t)$.
<pre>>> y2=x1-x2; >> plot(t,y2) >> ylim([-0.1 1.1]); >> legend('y2(t)')</pre>	

	Plot the output signal $y(t)$.
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2. System properties

In this section, the basic properties that a system can have will be presented. After the definition of each property, a typical example will be presented.

- Causal-non-causal systems.

A system is called causal when its output at time t , depends on the values of the input signal, for $\tau \leq t$. That is, for each input signal, the corresponding output depends only on the present and previous values of the input.

Thus in a causal system if the input is zero for a time interval

$$x(t) = 0 \text{ for } t \in [t_0, t_1]$$

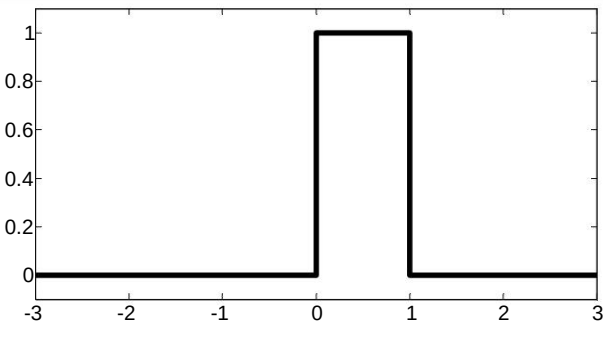
where t_0 constant number (i.e. $x(t) = 0$ for time t we get 0 as a result

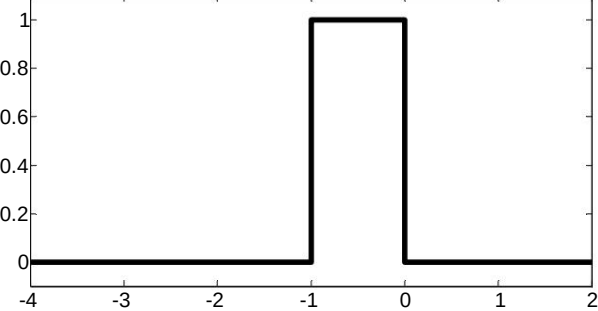
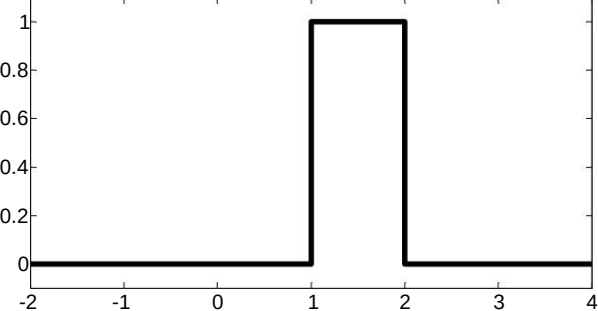
$$y(t) = 0$$

and the output is zero for the same time interval. (i.e. $y(t) = 0$ for time $t \in [t_0, t_1]$)

0

Example: Given a square wave input of duration 1s. Find whether the systems $y(t) = x(t)$ and $y(t) = x(t-1)$ described by the input-output relations $y(t) = x(t)$ and $y(t) = x(t-1)$ are causal.

COMMANDS	RESULT - COMMENTS
<pre>>> t1=-3:1:0; >> x1=zeros(size(t1)); >> t2=0:1:1; >> x2=ones(size(t2)); >> t3=1:1:3; >> x3=zeros(size(t3)); >> t=[t1 t2 t3]; >> x=[x1 x2 x3]; >> plot(t,x); >> ylim([-0.1 1.1]);</pre>	 Design of $x(t)$ (1) $y(t) = x(t)$ and $y(t) = x(t-1)$ with the method of drawing functions of many branches in space.

<pre>>> plot(t-1,x) >> ylim([-0.1 1.1]);</pre>	 Designing the output for the system with relation input-output $y(t) = x(t-1)$. We observe output appears in a time shorter from the one where the input appears. Therefore our system is non-causal.
<pre>>> plot(t+1,x) >> ylim([-0.1 1.1]);</pre>	 Designing the output for the system with relation input-output $y(t) = x(t+1)$. We observe output appears in a time greater than from the one where the input appears. Therefore our system is causal.

• Static (memoryless) and Dynamic (memory-based) Systems

A system is called a static or memoryless system if for each input signal the corresponding output, at any time, depends only on the value of the input at that time. If a system is not static, it is called a dynamic or memory-based system.

Example: Let us assume a square wave input of duration 1s. $x(t)$ Find if the systems described by the input-output relations

$y(t) = x(t)$

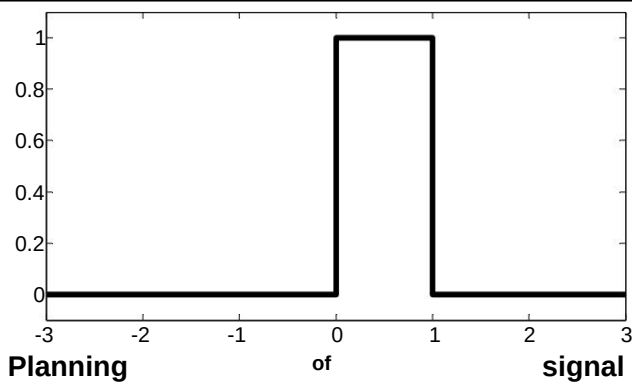
and

$y(t) = x(t-1)$

(1) are they static or dynamic:

COMMANDS	RESULT - COMMENTS
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```
>> t1=-3:1:0;
>> x1=zeros(size(t1));
>> t2=0:1:1;
>> x2=ones(size(t2));
>> t3=1:1:3;
>> x3=zeros(size(t3));
>> t=[t1 t2 t3];
>> x=[x1 x2 x3];
>> plot(t,x);
>> ylim([-0.1 1.1]);
```

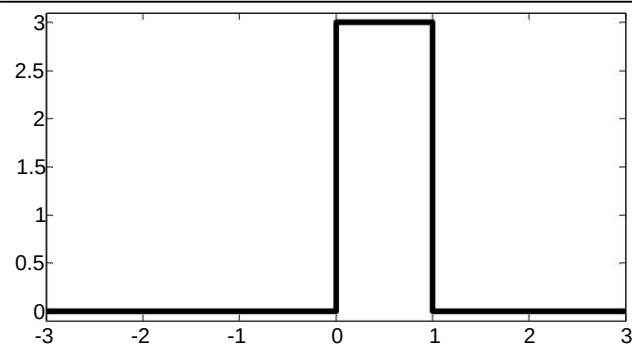


Planning of **signal**

by the method

designing multi-branch functions in space

```
>> plot(t,3*x);
>> ylim([-0.1 3.1]);
```



Designing the output for the system with relation

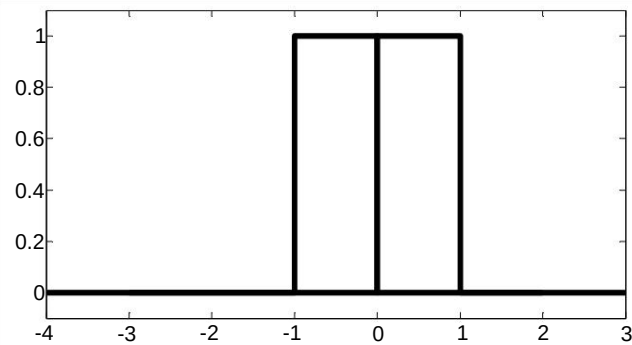
input-output $y(t) = 3x(t)$. The output depends

only from the price of entry to this

time t . Therefore our system is static or

amnesiac.

```
>> plot(t,x);
>> hold on?
>> plot(t-1,x);
>> ylim([-0.1 1.1]);
>> hold off?
```



Designing the output for the system with relation

input-output The value of output at

time t depends on the values

input at time t and $t-1$. Therefore the system

it has memory and is considered dynamic.

- Linear and nonlinear systems.

Let us consider a system whose output $y(t)$ is expressed as $y(t)$. Then a system is called linear if for any two input signals

$$x_1(t) \text{ and } x_2(t) \text{ the output is } y_1(t) \text{ and } y_2(t) \text{ respectively. Then for the input } a x_1(t) + b x_2(t) \text{ the output is } a y_1(t) + b y_2(t).$$

where a and b are scalar quantities and y expresses the system.

The above property generalizes to any linear combination of a finite number of input signals. This relation is often also known by the name superposition.

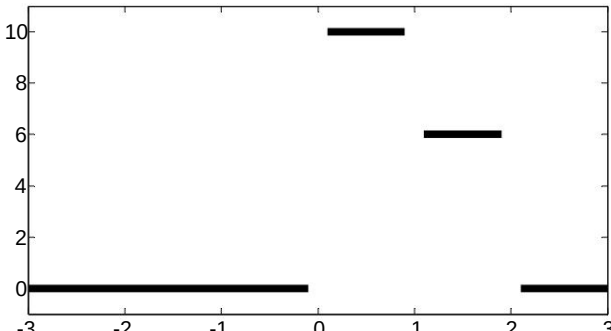
Example: Let there be two square pulse inputs, $x_1(t)$ and $x_2(t)$. Find whether the system is linear or not. (1)

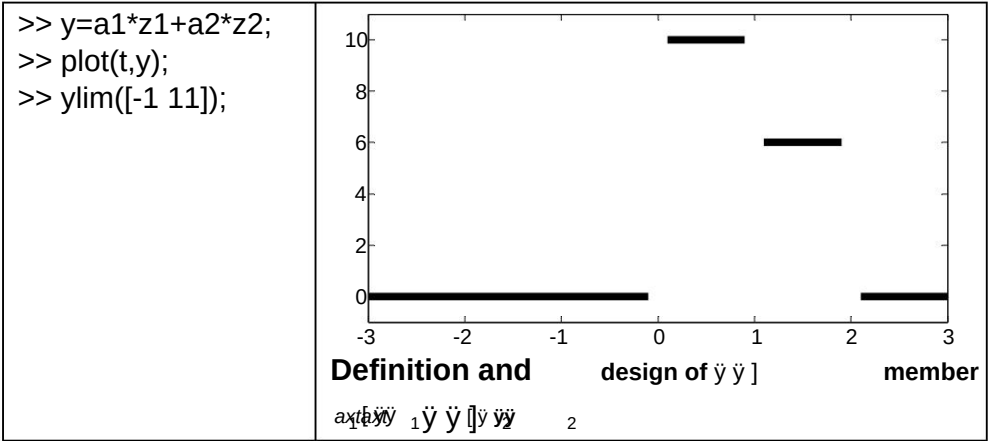
The system is described by the relations

input-output $y(t) = 2x_1(t) + 3x_2(t)$ are linear. (2) and $y(t) = 2x_1(t) + 3x_2(t)$

We will check the linearity relationship for interval $t \in [-3, 3]$ $a = 2$ and $b = 3$ in a timeline

For the system with input-output relationship $y(t) = 2x_1(t) + 3x_2(t)$ have we

COMMANDS	RESULT-COMMENTS
<pre>>> t=-3:1:3; >> x1=heaviside(t)-heaviside(t-1); >> x2=heaviside(t)-heaviside(t-2);</pre>	Definition of the input signals $x_1(t)$ and $x_2(t)$.
Calculation of the left-hand side of the relation	
<pre>>> a1=2; % a1=2 z=a1*x1+a2*x2; >></pre>	Definition of the relation $a_1 x_1 + a_2 x_2$ $a_2=3$; $>>$
<pre>y=2*z; >> plot(t,y); >> ylim([-1 11]);</pre>	 Definition and design of $y(t)$ member
Calculation of the right-hand side of the relation	
<pre>>> z1=2*x1; % 1 y y and y z2=2*x2;</pre>	Definition of $y(t)$ $y(t) = 2x_1(t) + 3x_2(t)$



We observe that the left and right members of the linearity relation are equal. Therefore, the system with input-output relation $y(t) = 2x(t)$ is linear.

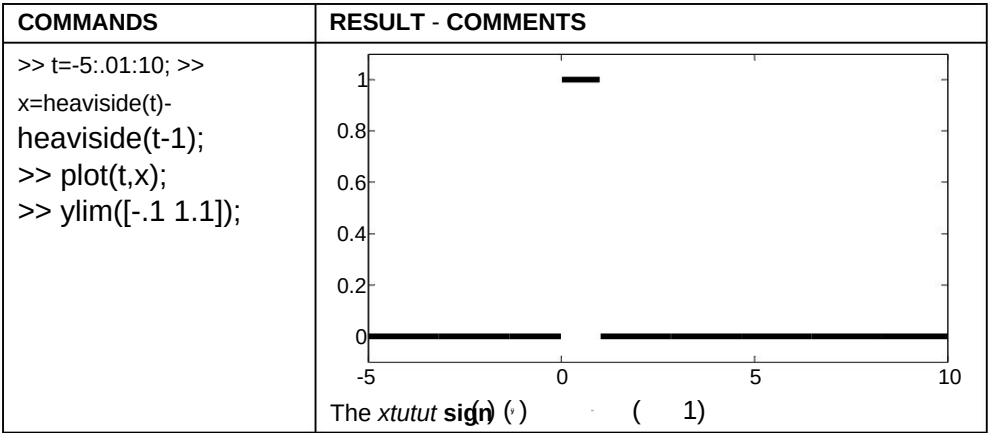
- Time-invariant and time-varying systems.

A system is said to be time invariant if and only if time shifts in the input signal translate into corresponding time shifts in the output. In such a system, if an input signal produces an output signal, then the input signal produces an output signal.

Example: Let us consider a system in which for an input $x(t)$ it gives $y(t)$ unchanged.

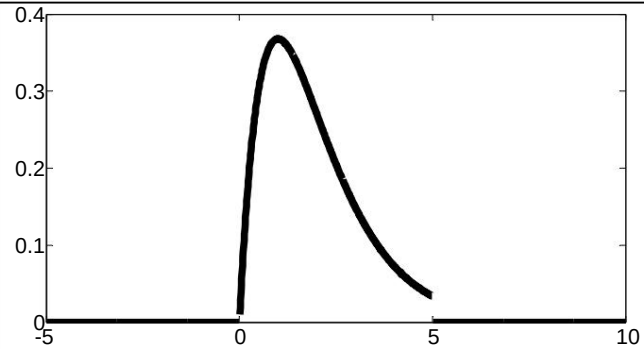
Find whether this system is temporally $y(t) = x(t-5)$ or $y(t) = x(t)$.

First we calculate the output of the system for an input shifted right by $t \rightarrow t-1$, that is, for an input signal $x(t-1)$. According to the output relation, replacing t with $t-1$ we will have $y(t-1) = x(t-1)$.



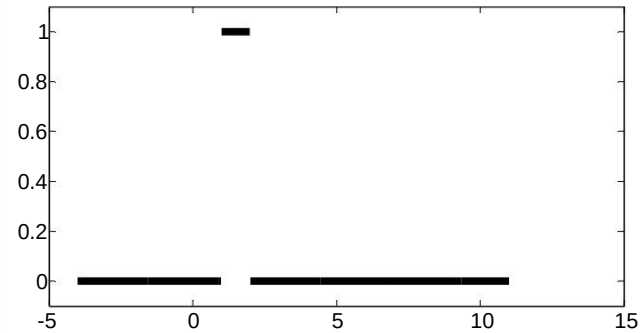
```
>>y1=t.*exp(-t).*(heaviside(t)-heaviside(t-5));

>> plot(t,y1);
```



The output signal $y(t)$ we have input signal $x(t)$,when

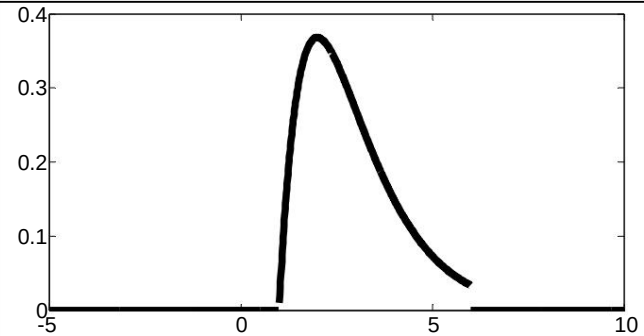
```
>> plot(t+1,x);
>> ylim([-1 1.1]);
```



The signal $x(t)$

```
>>y2=(t-1).*exp(-t+1).*(heaviside(t-1)-heaviside(t-6));

>> plot(t,y2);
```

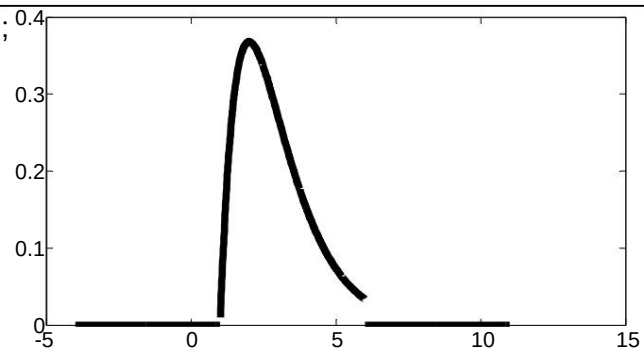


The output

(signal $y(t)$) where
($y(t)$ is obtained for the input signal $x(t)$)

Now we will plot the output signal displacement according to output signal $y(t)$ that we originally

```
plotted: >> plot(t+1,y1);
```



The output signal shifted by $t \rightarrow t+1$ right
 $y(t+1)$

We observe that the shifted output signal $y(t-1)$ coincides with the output that we got for the input signal $x(t-1)$. Therefore our system is time-dependent unchangeable.

• Reversible and irreversible systems

A system is said to be reversible when knowledge of the output makes it possible to calculate the input signal. More simply, we define a system as reversible if the relationship between input and output is one to one, that is, different input values give different output values.

Example: Find whether the systems described by the input relations $y_1(t) = 3x(t)$ and $y_2(t) = x(t)^2$ are reversible.

We will work by defining an input signal $x(t) = 2 - t$, $t \in [-2, 2]$.

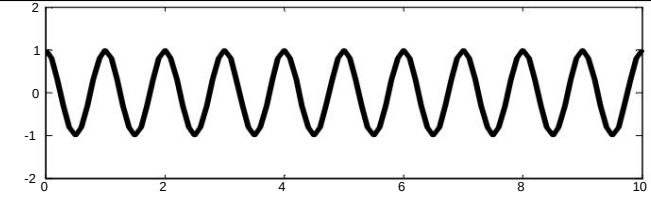
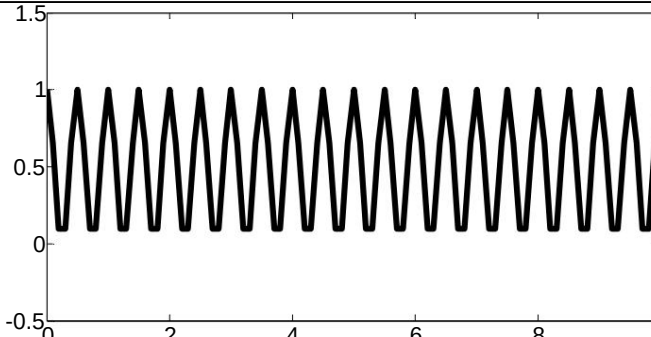
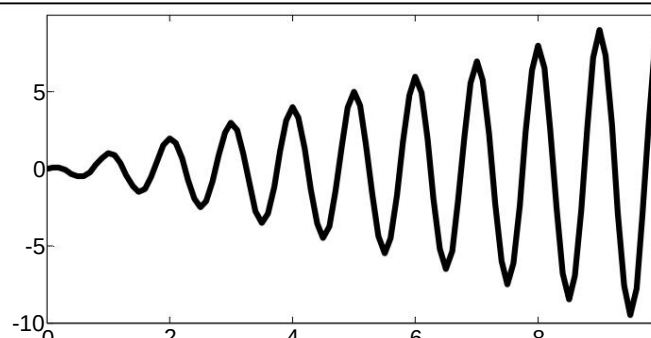
COMMANDS	RESULT	COMMENTS
>> t=-2:2; >> x=2*t	x = -4 -2 0 2 4	The input signal $x(t)$
>> y1=3*x	y1 = -12 -6 0 6 12	The output signal for the system $y_1(t) = 3x(t)$.
>> y2=x.^2	y2 = 16 4 0 4 16	The output signal for the system $y_2(t) = x(t)^2$.

We observe that the input-output relationship for the system described by $y_1(t) = 3x(t)$ is one-to-one so the system is reversible. The input-output relationship for the system described by the input-output relation $y_2(t) = x(t)^2$ is not one-to-one so the system is irreversible.

• Stable and unstable systems

Stability is a very important property of systems. In practice, it means that when a system has a small input, the output of the system does not diverge (goes infinite). A more formal definition is that a system is said to be stable or more correctly BIB stable (Bounded Input Bounded Output (BIBO) stability) when for every bounded input its output remains bounded.

Example: Let an input $x(t) = \cos(2t)$ excite the systems with input-output relations $y_1(t) = x(t)^2$ and $y_2(t) = tx(t)$. Find whether these two systems are stable.

COMMANDS	RESULT - COMMENTS
<pre>>> t=0:1:10; >> x=cos(2*pi*t); >> plot(t,x); >> ylim([-2 2]);</pre>	 Definition and plotting of the signal $x(t)$. We observe that $y_1(t) = x(t)^2$ is $x(t)^2 \leq 1$, that is, the signal is bounded for $M = 1$ since $x(t) \leq M$.
<pre>>> y1=x.^2; >> plot(t,y1); >> ylim([-0.5 1.5]);</pre>	 Definition and plotting of the signal $y_1(t)$. We observe that $0 \leq y_1(t) \leq 1$, that is, the signal is bounded for $M = 1$ since $y_1(t) \leq M$. Therefore the system with input-output relationship $y_1(t) = x(t)^2$ is BIBO stable.
<pre>>> y2=t.*x; >> plot(t,y2);</pre>	 Definition and plotting of the signal $y_2(t)$. We observe that as time increases the amplitude of the signal $y_2(t)$ increases as well. Therefore, the system with input-output relationship $y_2(t) = tx(t)$ is not BIBO stable.

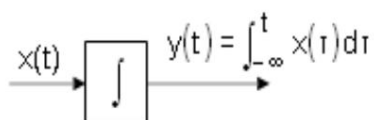
To be formally correct we should calculate the values of the signals when $t \rightarrow \infty$. This will be done with the help of the limit function.

COMMANDS	RESULT	COMMENTS
<pre>>> syms t >> x=cos(2*pi*t); >> limit(x,t,inf)</pre>	<pre>answer = -1 .. 1</pre>	We calculate $\lim_{t \rightarrow \infty} x(t)$ and we see that it ranges from interval $[-1 \ 1]$. Therefore the input $x(t)$ is blocked
<pre>>> y1=x^2; >> limit(y1,t,inf)</pre>	<pre>answer = 0 .. 1</pre>	We calculate $\lim_{t \rightarrow \infty} y_1(t)$ and we see that it fluctuates in the interval $[0 \ 1]$. Therefore the output $y_1(t)$ is blocked and the system is FEFE stable. The
<pre>>> y2=t*x; >> limit(y2,t,inf)</pre>	<pre>answer = NaN</pre>	$\lim_{t \rightarrow \infty} y_2(t)$ does not exist, therefore the output $y_2(t)$ is not blocked and the system is unstable.

PRACTICE EXERCISES

ASK

1) Let the integrator system be shown in the figure below.



Such a system accepts as input a signal $x(t)$ and the output $y(t)$ is given by the relationship:

$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

You are asked to study this system and answer the following questions regarding its

- properties: a. Linear or nonlinear. b. Dynamic or static. c. Causal or non-causal. d. Time-invariant or time-varying. e. Stable or unstable.

ASK 2)

Examine the stability of the system with the input-output relationship

output: $y(t)$ $x(t)$

ASK 3)

To examine whether the system governed by the input-output relationship $y(t) = \frac{1}{4}x(t)$ is causal. ()